CAD VR: Guarded Evidence Control for Fast and Slow Agentic Video Retrieval

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Abstract. Video event retrieval requires locating precise moments from multimodal descriptions under time constraints. Embedding scores and agent confidence alone do not establish whether the evidence is sufficient to stop searching. We introduce CAD VR, which couples fast multimodal retrieval with slower tool-using agents through a provenance-aware evidence board, typed verification and action selection, and a code-level stop guard. We evaluate six configurations on 86 AIC 2026 TEST queries and conduct a subsequent live ablation of final verification. In that ablation, CAD VR achieves 58.1% strict R@1, versus 47.7% for always-two agents and 50.0% for always-two agents with final verification; its advantage over the latter remains inconclusive (exact McNemar $p = 0.092$). Relative to always-two agents, median latency increases from 53.7 to 75.8 s with negligible agent-call reduction. These findings show how to make retrieval decisions auditable, while the accuracy–latency trade-off motivates improved stopping policies and manual–agent cooperation.

Keywords: Video event retrieval · LLM agents · Fast and slow control · Decision models · Evidence verification · Multimodal retrieval

1 Introduction

Event retrieval maps descriptions to a known moment (KIS), a moment and answer (QA), or an ordered event chain (TRAKE). Joint embeddings generate candidates quickly [5,28], but exact localisation remains difficult: queries combine actions, appearance, text and speech, while broadcasts repeat similar scenes. On our benchmark, retrieval ranks the correct video first on 45.3% of queries, yet strict top-1 accuracy is only 15.1%.

Tool-using agents search, inspect frames and refine queries [24, 27, 39, 42]. Running both agents costs tens of seconds; their confidence is uncalibrated, and disagreement alone does not justify further search.

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CAD VR makes escalation, evidence assessment and stopping explicit and auditable through fast and slow control [20, 26]: System-1 retrieves, a non-generative decision model [36] assesses evidence and selects typed actions, and System-2 agents search and inspect images. "Fast and slow" denotes these roles, not lower end-to-end latency. Our contributions are:

- **Typed evidence-guided control**: a bounded controller separates candidate generation from two soft-role agents sharing ten Model Context Protocol (MCP) tools [1] in an operator-facing multimodal retrieval system.
- **Evidence board and stop guard**: provenance, typed support assessments and code-level rules expose abstention, escalation and budget termination in auditable traces.
- **Controlled evaluation and diagnostics**: a live C/C+V/E ablation and same-evidence pre/post verification contrasts within C+V distinguish final-ranking effects from the broader controller package. Independent hint prefixes and group-level metrics complement this component analysis.

2 Related Work

Interactive video retrieval. VBS systems combine embeddings and browsing [28, 33], spatial and temporal queries [8, 19, 21], answer grounding [35], fusion [22] and graded autonomy [15]. AIC systems use dynamic programming and visual verification [14, 30]; agents mediate video and lifelog search [16, 18, 37]. CAD VR adds typed control and an evidence guard above search agents.

LLM agents for video. Agents inspect retrieved frames [39], select tools [42], separate planning, grounding and verification [27], plan temporal scope [24] and coordinate retrieval [40]. We separate this interleaved reasoning and acting [41] from typed control and code-level stopping.

Fast and slow control and routing. SwiftSage pairs fast policies with slow planning [26]; FrugalGPT and RouteLLM trade model cost for quality [7, 31]. Unlike free-form judge rationales [44], CAD VR uses typed decision probabilities and a separate guard, with calibration diagnostics [13].

3 System Overview

The system (Fig. 1) indexes 1,478 broadcasts plus batch-two news, 298 traffic-camera and 12 cycling recordings, with organiser (382,299 frames) and InfoShot++ (2,212,383 frames) profiles. Our InfoShot++ sampling algorithm extends InfoShot [43] with multiple unusual within-shot candidates, native-frame-rate refinement around text/motion changes, and temporal gap filling. Its incremental benefit is not separately ablated. Milvus [38] stores PE-Core [5], Qwen3-VL [25] and GLAP [11] vectors; Elasticsearch stores OCR [17], WhisperX-aligned Vietnamese ASR [4, 23], audio tags [10] and frame/time mappings. TARA [3] indexes

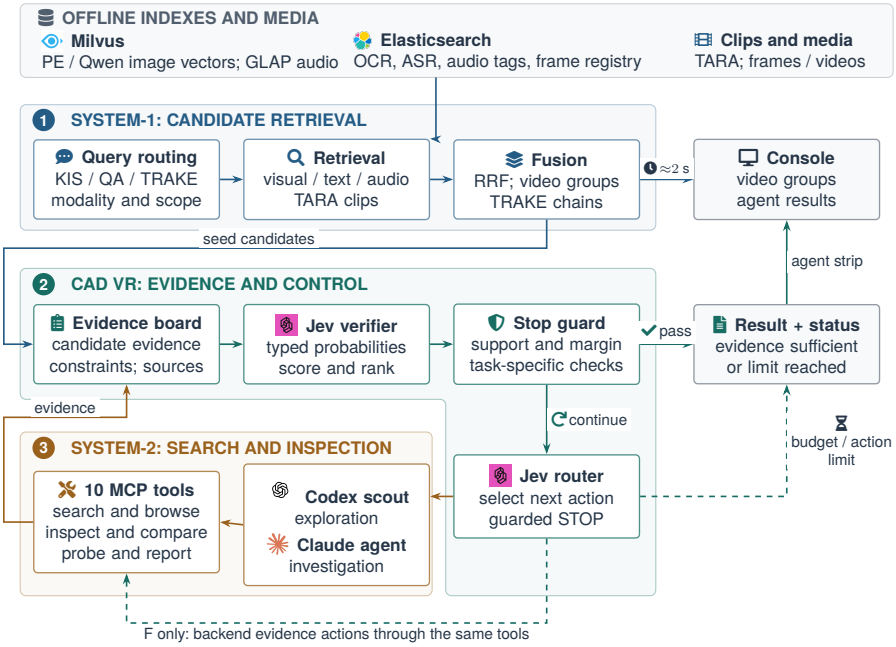


Fig. 1. CAD VR architecture: retrieval, evidence verification, guarded stopping and agent/tool actions. Dashed paths denote F-only macros and limit termination; evidence returns to the query-local board. Rankings stream to the console. Logos identify providers.

496,155 clips across core and batch-two videos (partial cycling coverage); traffic windows span 4/8/16 s. Retrieval returns groups in about two seconds; a sidecar reuses them and streams agent rankings. New searches cancel that run. Section 6 specifies the narrower benchmark configuration.

4 System-1 Retrieval and Group-by-Video Ranking

A heuristic parser routes each query to visual (English translation), OCR (exact and fuzzy Vietnamese phrases), speech, audio and clip channels, and infers folder-scope filters from the topic. Channel c returns a frame ranking L_c, and weighted reciprocal rank fusion [9] scores frame f by $R(f) = \sum_{c:f\in L_c} w_c/(60 + r_c(f))$ with ranks $r_c(f) \geq 1$; cosine scores from different embedding spaces are never added.

Group-by-video ranking. The console groups frames for playback (Fig. 2). For video v, returned frames F_v, best frame $f_v^\star$ and $R^\star = \max_f R(f)$,

$$S(v) = 0.75\,\frac{R(f_v^\star)}{R^\star} + 0.15\,\frac{\text{mean}_{f\in\text{top3}(F_v)}\,R(f)}{R^\star} + 0.10\,\min\Big(1, \frac{n_{30}(v)-1}{4}\Big), \quad (1)$$

where $n_{30}(v)$ counts returned frames within 30 s of $f_v^\star$, including itself. These heuristic weights are unablated; top-frame gaps over 30 s flag ambiguity. With



Fig. 2. Console workflow: query and search controls (left), grouped video results with scores and channels (centre), and frame inspection, evidence and submission identifiers (right).

TARA, video-level RRF combines channel rankings and one vote per clip scale, replacing Eq. (1).

TRAKE. A shared context (subject, clothing, scene) is composed with every event description, events are retrieved separately, temporal non-maximum suppression keeps distinct moments, and an exact dynamic program selects the maximum-weight chain with increasing times in each video. Videos are ranked by event coverage, then chain quality.

5 CAD VR: Guarded Evidence Control

Evidence board. Each query owns a fresh board $\mathcal{B}_t$ of constraints, candidates and evidence; no previous hint or reasoning enters it. A deterministic compiler retains the query (weight 2), extracts clauses and attribute, relation or negation spans (weight 1), and adds TRAKE events (weight 1.5) or a QA answer constraint (weight 2). Evidence is de-duplicated by kind and source. An agent's report counts as a visual observation only after that agent loaded the frame; this records inspection, not guaranteed truth. Agent confidences order that agent's proposals but are not used as probabilities.

Typed verification. Each step verifies a shortlist of five candidates, prioritising each agent's leading proposal and the verified leader, then distinct moments from the fused ranking. TRAKE can expand it to 12 to cover the leading video's events. For candidate v and constraint c, Jev 1.13 returns a typed distribution over $\{$*supported*, *refuted*, *unknown*$\}$, with instructions not to infer visual details from embedding scores or confidence. Here "non-generative" describes the decision interface: it returns probabilities rather than free-form plans or search queries.

Algorithm 1 CAD VR controller for one query (6 steps, 40 tool calls, 240 s).

Input: query q, task type, System-1 result R_0

$\mathcal{C}_q \leftarrow \textsc{Compile}(q)$; $\mathcal{B}_0 \leftarrow \textsc{Seed}(R_0, \mathcal{C}_q)$; $\mathcal{I} \leftarrow \emptyset$ ▷ fresh, query-local
for $t = 0, 1, \ldots$ **while** budget remains **do**
  $\textsc{JevVerify}(\mathcal{B}_t)$; $\textsc{Publish}$(ranking of $\mathcal{B}_t$)
  if guard (3) holds **then return** ▷ evidence sufficient
  $\mathcal{A}_t \leftarrow \{\texttt{STOP}\} \cup \{\text{calls to agents} \notin \mathcal{I}\} \cup \{\text{untried evidence actions in mode F}\}$
  $a_t \leftarrow \textsc{JevRoute}(\mathcal{B}_t, \mathcal{A}_t)$; **if** $a_t = \texttt{STOP}$ **then** $a_t \leftarrow$ next escalation in $\mathcal{A}_t$
  if $a_t = \texttt{STOP}$ **then return** ▷ actions exhausted
  $\mathcal{B}_{t+1} \leftarrow \textsc{Execute}(a_t, \mathcal{B}_t)$; $\mathcal{I} \leftarrow \mathcal{I} \cup$ agents in a_t

Without evidence, code sets $p^{\text{sup}} = 0$. It aggregates

$$s(v) = \exp\Big(\sum_{c \in \mathcal{C}(v)} w_c \log \max\big(\epsilon, p_c^{\text{sup}}(v)\big) \Big/ \sum_{c \in \mathcal{C}(v)} w_c \Big), \tag{2}$$

a weighted geometric mean that penalises weakly supported constraints; the separate guard requires every applicable constraint to pass. *Holistic* mode (E) verifies the query and answer constraints; *constraint-level* mode (F) verifies all applicable constraints and enables evidence actions. Ranking promotes candidates for which *supported* is the largest probability on every assessed constraint, demotes those with any *refuted* maximum, and otherwise preserves fused System-1/agent order. Thus *unknown* is abstention; promotion need not satisfy the stricter stop threshold.

Routing and the stop guard. While the guard fails, Jev chooses the next action from $\mathcal{A}_t \subseteq \{$`STOP`, `CALL_CODEX`, `CALL_CLAUDE`, `CALL_BOTH`, `SEARCH_MORE`, `INSPECT_TOP1`, `COMPARE_TOP2`$\}$, each described in one sentence; the first request also classifies modality and query structure. Each agent runs at most once. The evidence actions (mode F only) are backend macros: a focused search for the least-supported constraint, or OCR and speech around the top one or two candidates. Agent disagreement enters the state as a signal, and the router is told that it alone does not justify another agent. Evidence-sufficient stopping requires, for the top candidate v_1,

$$\begin{gathered} s(v_1) \geq \tau, \qquad s(v_1) - \max_{v' \in \text{comp}(v_1)} s(v') \geq \delta, \\ p_c^{\text{sup}}(v_1) > \max\big(\tfrac{1}{2},\ p_c^{\text{ref}}(v_1),\ p_c^{\text{unk}}(v_1)\big)\ \forall c, \end{gathered} \tag{3}$$

where competitors exclude the same moment (≤ 8 s apart in one video); TRAKE compares other videos. Only scored competitors enter the maximum, which defaults to zero. QA needs an answer; TRAKE needs supported events in increasing time. We use fixed raw-score thresholds $\tau = 0.9$, $\delta = 0.1$. A rejected `STOP` triggers the next available escalation (Codex, Claude, inspect, search, compare). Budget or action exhaustion still returns the current ranking, marked separately from evidence-sufficient stopping (Algorithm 1). Passing the guard is a policy decision, not a correctness guarantee.

System-2 agents and tools. Codex CLI with GPT-6.1 Sol [32] and Claude Code with Opus 5.5 [2] use high reasoning effort and fast mode in fresh processes without shell/file tools. Both share ten MCP tools for retrieval, corpus browsing, local frame/text inspection, candidate comparison and reporting (inventory and call counts in the supplement). Soft roles favour exploration or counterexample inspection without restricting capabilities. Prompts describe the corpus and include a bounded board snapshot marked as untrusted data.

6 Experimental Setup

Benchmark. We assembled 111 queries from the organisers' AIC 2026 query workbooks and final-round submissions. Ground truth combines DRES-accepted submissions [34] with owner-reviewed annotations, each checked against the source frames or clips: moments for all tasks, accepted answers for QA and ordered per-event alternatives for TRAKE. Queries are split by content hash and grouped by target video, so no target video appears in both DEV (25) and TEST (86). TEST has 59 T-KIS, 19 QA and 8 TRAKE queries over 87 target videos, using InfoShot++ (46 queries) or organiser (40) profiles. DEV fixed the policy; its fitted temperature calibration was not adopted. All configurations use raw probabilities.

Configurations. The initial six share System-1, the soft roles and the ten tools. **A**: retrieval only. **B**: always Codex. **C**: Codex and Claude in parallel after retrieval. **D**: Jev holistic reranking without agents. **E**: CAD VR with holistic verification. **F**: CAD VR with constraint-level verification and evidence actions. B and C fuse retrieval and agent proposals by RRF. System-1 uses PE-Core for visual search on both profiles, heuristic parsing with VI→EN translation, no LLM rewriting or reranking, and the top 150 frames in at most 30 videos; routing activated OCR in 18, speech in 11, audio in 9 and TARA in 5 TEST queries.

Metrics. Strict R@k and MRR require a moment within ± 1 s; QA additionally requires a normalised accepted answer, and TRAKE a complete ordered sequence in one video. *Video* R@k checks target identity among the first k distinct videos. *Group-moment* R@k checks whether those videos contain a returned matching moment, ignoring QA answers but retaining TRAKE's complete-sequence requirement. Video order follows first occurrence in the saved ranking. Rescue@1 measures strict success with agent provenance among runs whose own retrieval seed failed. We record latency, calls and tokens, plus ECE (10 bins) and Brier score [6] over final scored candidates of non-TRAKE queries. Reliability plots use five bins; candidate observations are not independent query samples.

Protocol. Each query–configuration pair is a fresh live run, with variants in seeded random order per query. The initial comparison retains 516 completed TEST runs. The supplementary C/C+V/E round uses the same 86 queries (258 runs), raw policy, prompts, tools, retrieval and budgets. C+V adds E's final holistic

Table 1. Initial TEST round (N=86; tolerance 1 s). R@1: bootstrap 95% CI, 2,000 draws; Resc.: Rescue@1, 73 eligible runs/variant; p50: median latency; Agents: invocations/query; kTok: mean agent and decision tokens in thousands.

	Configuration	R@1 [95% CI]	R@5	MRR	Resc.	p50 (s)	Agents	kTok
A	Retrieval only	15.1 [8.1, 23.3]	33.7	.233	–	1.6	0	0
D	Jev rerank	15.1 [8.1, 23.3]	33.7	.233	0.0	2.1	0	3
B	Always Codex	39.5 [29.1, 50.0]	68.6	.529	28.8	50.5	1.00	230
C	Always both	51.2 [40.7, 61.6]	**72.1**	.612	42.5	51.2	2.00	323
E	CAD VR, holistic	**58.1** [47.7, 68.6]	67.4	**.630**	**52.1**	72.6	1.95	316
F	CAD VR, constraint	52.3 [41.9, 62.8]	64.0	.572	45.2	84.2	1.99	416

verifier to C, with no routing or adaptive stopping, and saves pre-verification rankings. This separate round reuses analysed TEST data without new DEV tuning.

Quota-interrupted attempts are archived and retried from fresh state. The supplementary round has no completed-run errors/fallback. Its 13 interruptions add 314.6 s and 25 agent calls, excluding pauses; tables report completed-attempt latency. Percentile bootstrap intervals [12] resample base queries: 2,000 draws for initial per-configuration intervals and 10,000 paired draws (seed 2026) for contrasts. Exact McNemar tests [29] complement these exploratory, multiplicity-uncorrected comparisons. Code, manifests, configuration hashes and query-level reports are available in the benchmark repository[3]. Competition media are not redistributed.

7 Results

7.1 Initial Six-Configuration Comparison

E has the highest observed strict R@1 in Table 1 (58.1%). E improves on B by 18.6 points (95% CI [9.3, 27.9]; McNemar $p < 0.001$). Its E–C difference is inconclusive (+7.0, [−2.3, 16.3]; 11 E-only and 5 C-only successes, $p = 0.21$). C has the highest R@5 (72.1% vs. 67.4%).

D equals A on retrieval accuracy metrics: all 525 assessments were *unknown*, preserving System-1 order. This shows no gain from verification of retrieval-only evidence in this setting; it does not isolate verification after agent inspection. F does not improve overall on E (−5.8 points, [−15.1, 3.5]), and changes both verification granularity and available evidence actions.

F leads initial TRAKE R@1 (50.0% versus E's 37.5% and C's 25.0%), but eight queries make this suggestive only.

7.2 Live C/C+V/E Ablation

Table 2 tests final verification after always-two acquisition. All 86 C+V traces show both agents, one final verification and no routing. C+V–C and E–C+V

[3] https://github.com/vonhatbao2205/AIC2026/tree/main/benchmarks/agent

Table 2. Supplementary TEST round (N=86); recall in percent. Lower panel: paired R@1 contrasts, 95% bootstrap CIs (10,000 draws), exact unadjusted McNemar p and discordant win/loss counts. "Pre": C+V's own ranking before verification.

	R@1	R@5	MRR	p50 (s)	p95 (s)	Agents	Jev calls
C	47.7	**69.8**	.579	53.7	103.2	2.00	0.00
C+V	50.0	66.3	.581	55.6	98.4	2.00	1.00
E	**58.1**	66.3	**.620**	75.8	198.9	1.95	4.91

Contrast	ΔR@1 (pp) [95% CI]	Win/loss	p
C+V − C	+2.3 [−3.5, 9.3]	5/3	.727
E − C+V	+8.1 [0.0, 16.3]	10/3	.092
E − C	+10.5 [2.3, 18.6]	12/3	.035
C+V − Pre	+3.5 [−1.2, 9.3]	4/1	.375

remain inconclusive. E–C is positive in this round (unadjusted $p = 0.035$), unlike the initial round, but does not establish a gain over the C+V control.

C/C+V use identical acquisition policies but independent trajectories. E–C+V is a controller-package contrast: scheduling, intermediate verification, board-conditioned context and stopping vary together. By task, C/C+V/E obtain 57.6/61.0/74.6% T-KIS, 26.3/26.3/31.6% QA and 25.0/25.0/0.0% TRAKE R@1. E's unchanged aggregate 58.1% across rounds therefore hides task-level variability, including the loss of its three initial TRAKE successes.

7.3 What Does Final Verification Contribute?

On identical C+V candidates and answers, final verification raises R@1 from 40/86 (46.5%) to 43/86 (50.0%): four repairs and one harm. This same-evidence difference is inconclusive (Table 2).

An initial-round offline diagnostic orders final candidates by logged RRF, first-seen time and frame ID, reproducing all 344 A/B/C/D rankings. E's RRF yields 51/86 versus verified 50/86 (one repair, two harms); F yields 48/86 versus 45/86 (one repair, four harms). Paired differences are inconclusive (95% CIs $[-5.8, 2.3]$ and $[-9.3, 1.2]$ points). This limits general reranking claims; earlier verification may affect acquisition and context.

7.4 Accuracy, Latency and Computation

In the initial round, CAD VR did not save computation (Fig. 3): E invoked 1.95 agents per query, and its median latency (72.6 s; p95 171 s) exceeds C's (51.2 s; p95 94 s). The router sent the investigator first in 83/86 queries; sequential escalation adds agent times. Jev averaged 4.9 calls and 3.2 s/query (Fig. 4a). The supplementary round confirms this pattern (Table 2); E/C+V/C use 327k/344k/346k tokens. Calls and tokens are workload proxies, not FLOPs or complete financial cost.

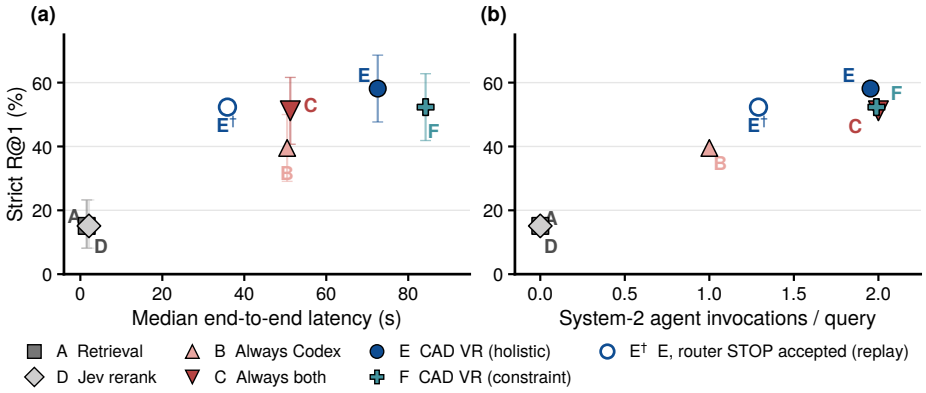


Fig. 3. Initial-round strict R@1 versus latency (a; 95% bootstrap CIs) and agent calls (b). E† is an offline first-`STOP` replay, not a deployed configuration (Sect. 7.5).

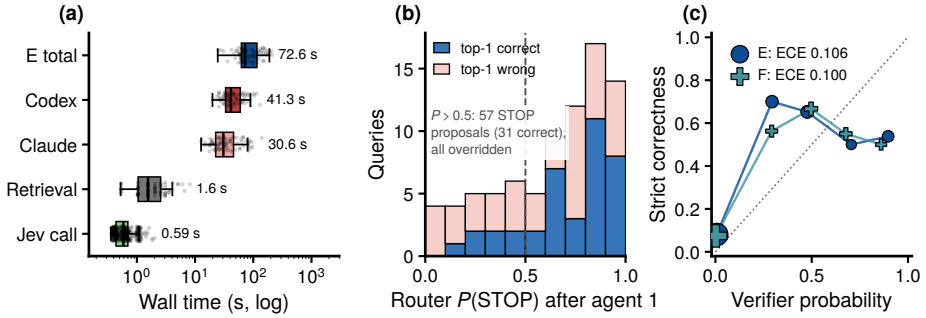


Fig. 4. Initial controller diagnostics. (a) E wall time per layer (log scale). (b) Router P(STOP) after agent 1 by top-1 correctness; every proposal above 0.5 was overridden. (c) E/F verifier reliability (marker area $\propto$ count).

7.5 What Controls Compute: Router versus Guard

After the first agent, the initial E router proposed `STOP` with probability above 0.5 in 57/82 decisions (median 0.73): 31/57 top-1 answers were correct, versus 7/25 otherwise (Fig. 4b). The guard blocked all 57; only nine runs ended evidence-sufficient. Offline E† retains the ranking, calls and elapsed time immediately before the first router `STOP`, or the final outcome if none occurs: 1.29 agents/query, 35.9 s median and 52.3% R@1, versus E's 1.95, 72.6 s and 58.1%. This post-hoc TEST replay fixes earlier decisions; it measures neither live latency nor equivalence to C. Deployment requires DEV selection and new live evaluation.

Verifier scores imperfectly predict strict correctness (ECE 0.106, Brier 0.112; Fig. 4c): only 7/13 candidates scoring at least 0.8 were correct. Labels concern exact moments/answers, not human-assessed constraint support; neither scores nor guard terms guarantee correctness.

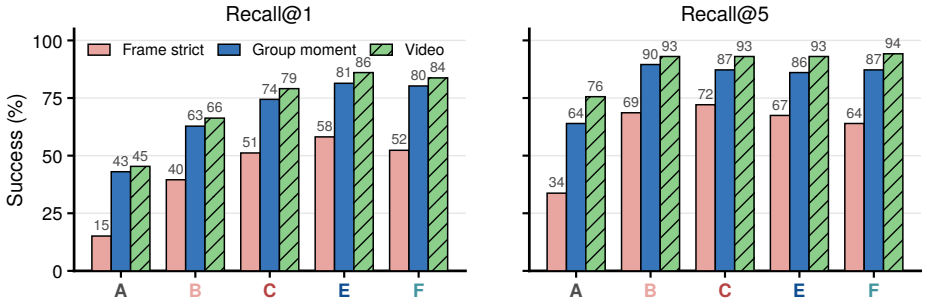


Fig. 5. Initial strict, group-moment and video R@k (D=A). Group-moment checks evidence in the first k videos, ignoring QA answers but requiring complete TRAKE sequences. This compares metrics, not grouping algorithms.

7.6 Group-by-Video Evaluation

Figure 5 separates exact-answer accuracy from coverage in the returned video groups. Retrieval has strict, group-moment and video R@1 of 15.1%, 43.0% and 45.3%; E reaches 58.1%, 81.4% and 86.0%. Group-moment R@5 is 86–90% for the agent configurations. These same-ranking metric gaps motivate browsing, not a claimed gain over ungrouped retrieval. QA's 89.5% group-moment versus 42.1% strict R@1 for E combines two effects, selecting the exact frame and supplying an accepted answer. Neither answer extraction alone nor operator effort is isolated.

7.7 Cumulative-Hint Evaluation

Twenty-four TEST queries (12 T-KIS, 12 QA) have three or four reviewed hints whose concatenation equals the full query. Each prefix h_1, $h_1{+}h_2$, … is an *independent* query run from a fresh state, without cross-hint memory or any progressive mechanism; we compare A, C and F. With only the first hint (Fig. 6a), F ranks the target video first for 83.3% of queries, against 66.7% for C (+16.7 points, $[4.2, 33.3]$) and 41.7% for A. On the full query, strict R@1 is 41.7%, 33.3% and 12.5% (F vs. A +29.2, $[12.5, 50.0]$; F vs. C +8.3, $[-8.3, 25.0]$). F uses 1.79/2.0 agents and 9.2/12.3 Jev calls at first/full hint. Its video R@1 succeeds early on 23/24 queries (95.8%; Table 3). First success is retrospective: later prefixes may fail, precluding online-stopping or compute-saving claims.

7.8 Qualitative Example

Figure 7 traces an initial-round TEST query through E. Retrieval finds the right video but an interview scene. Claude finds the feeding moment; Jev supports it (0.73) and refutes a near-duplicate (0.81). The router proposes `STOP` (0.97), but the guard rejects it ($0.73 < \tau$), triggering Codex's local inspection.

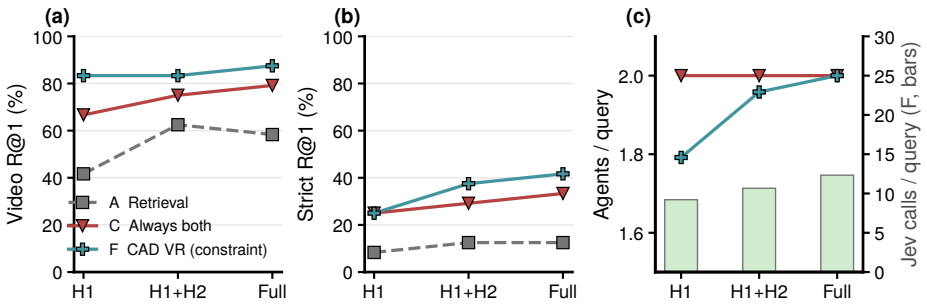


Fig. 6. Cumulative-hint evaluation on 24 TEST queries, each prefix a fresh, independent run. (a) Video R@1. (b) Strict R@1. (c) Agents per query (lines) and Jev calls per query for F (bars).

Table 3. Hints-to-solve on 24 queries. $h^\star$: first successful prefix (H+1 if unsolved); Early: success before the last hint; Hint saving: $(H - h^\star)/(H - 1)$, 0 if unsolved. These retrospective measures do not imply online stopping or compute savings.

	Video criterion				Strict criterion			
	Solved	Mean $h^\star \downarrow$	Early	Hint saving	Solved	Mean $h^\star \downarrow$	Early	Hint saving
A	19	2.25	.750	.611	6	4.04	.250	.153
C	22	1.67	.875	.792	10	3.42	.375	.319
F	**23**	**1.38**	**.958**	**.889**	**12**	**3.25**	**.458**	**.361**



Fig. 7. Initial-round TEST example. Top: top-1 frames of retrieval (A), always-two (C) and CAD VR (E). Bottom: E's logged controller events (teal) and agent work (amber). The left frame's lower-third is blurred.

8 Best Practices for Time-Constrained Search

Search manually while agents run on the full query in a separate tab; disable agents in the manual tab to avoid cancellation. Inspect streamed candidates and frame/OCR/ASR evidence immediately. These practices lack controlled operator timing; LLM rewriting is outside the benchmark.

T-KIS: select clip retrieval. Preserve action relations in keywords or rewrites. In `tkis-query-23`, a worker towing a bin by motorcycle drops a white bag. TARA returned the correct video and segment first from hint 1; PE failed in that competition attempt. Verify with later hints (`N061-V002`, 289.607 s). This is an operator observation, not a matched ablation or measured speedup.

T-KIS: use concurrent agents. In `tkis-query-10`, cyclists pass white balloons onto a bridge, then six riders appear frontally. Manual PE/TARA localisation was difficult; a concurrent agent found `S01-V009`, 4542.7 s. Independently, initial A/C rank it second/first; C reports it at 30.4 s, finishing at 62.5 s. Submission time was unmeasured.

TRAKE: align events manually. Retain context and event conditions; use agent windows when stalled. The fish-hotpot sequence in `L26_V402` is 91.24, 140.48, 189.12 and 224.24 s. Inspect first-contact/last-entry boundaries on paused frames; increasing DP chains cannot guarantee alignment. Independent exact first-frame labels are unavailable. Check event count/order and identifiers. Listed times are retrospective, not search inputs.

9 Discussion and Limitations

Attribution and computation. E–C+V establishes neither superiority nor equivalence; scheduling remains unisolated. Revised stopping requires live evaluation.

Validity. Corpus size/language and owner review limit generalisation and annotation independence. Video-disjoint splits retain thematic overlap; bootstraps omit shared-video dependence. Two rounds cannot reliably estimate C/E variability; C+V runs once/query on previously analysed TEST data. Service identifiers cannot freeze hosted behaviour. F changes granularity and actions. Text-only verification inherits observation errors; geometric support is uncalibrated for joint correctness. Group weights are unablated; operator effort is unmeasured.

10 Conclusion

CAD VR makes escalation and stopping auditable. E leads observed R@1, with inconclusive E–C+V gains, mixed ranking effects and no compute savings. Diagnostics motivate human verification; stronger causal or efficiency claims require repeated evaluation and development-selected stopping.

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